Audit: errors, validation bias, and the role of Monte Carlo and scoring

Date: 14 October 2026 · **Scope:** everything produced so far (code, acceptance protocol, PDF report v2.0, chat summaries) **Reproduce:** `python run_engine.py && python audit/run_audit.py` → `results/engine/`, `results/audit/`

The question behind this audit: *is every number we reported an objective, reproducible estimate of how the model would perform on new slopes?* Each suspected bias was **measured**, not argued. Where an earlier claim was wrong or inflated, it is corrected below and in the code, the acceptance protocol and the PDF (v2.1).

1. Bottom line

	Before audit (v2.0)	After audit (v2.1)
Hybrid engine AUC	0.903 (soil domain) / 0.906	**0.860** on all cases (CI 0.832–0.896)
Comparison baseline	"AHP-like" index, 0.68	Equal-weight score (**not AHP**) 0.581; **fair baseline: logistic regression 0.731**
Acceptance	16/17 pass	**16/19 pass** (4 are design checks), **1 FAIL**, **2 PENDING**
Physics P_f	"a probability usable in risk assessment"	**Not calibrated** on real slopes (ECE 0.33)
Field score tool (HI/RI)	Implied validated	**Not validated** against any landslide outcome
Sobol' weights	"objective, replace AHP weights"	Depend on class definitions; judgement **moved**, not removed

What still holds:

- The physics is solved correctly.
- The Monte Carlo is numerically stable.
- On real case histories, a physics-constrained learner clearly beats every linear or score-based alternative.
- Its probabilities are reasonably calibrated (ECE 0.077).

2. Mistakes and overclaims found

#	Earlier claim	What was wrong	Evidence	Fix
1	Hybrid AUC 0.903 / 0.906	Three optimistic choices stacked: (a) soil domain chosen **after** seeing results; (b) AUC computed on predictions **averaged over 5 repeats**, an ensemble effect; (c) γ and H were **unconstrained**, so the model learned jagged, non-physical responses to them	(b) +0.005 to +0.012. (c) AUC fell from 0.900 to 0.862 once γ, H were constrained. The partial dependence on H jumped −1.0 → +1.7 → −0.5 log-odds between nearby values, a memorisation signature	Primary analysis = all cases; single-pass estimator; γ and H constrained to their physical direction
2	Hybrid "clearly beats an AHP-style index"	Baseline A was an **equal-weight quintile score I designed**, with assumed directions. No AHP weights were tested. It was a strawman.	A = 0.581; logistic regression on the same 6 inputs = 0.731	Renamed A; added fair baseline E. Hybrid beats E by **+0.132** (group-bootstrap CI 0.093–0.172)
3	"16 of 17 acceptance criteria pass"	4 passes are **true by design** (N1 follows from N; old A7 is guaranteed by monotone constraints). Most thresholds were **set by me**, not taken from standards. The calibration criterion was switched to the most favourable ECE estimator **after** the per-fold estimate failed (0.12 > 0.10).	Per-fold 0.116; single-pass 0.077; 5-repeat averaged 0.071	Each threshold now labelled *literature / project-set / analytic*. ECE uses the single-pass estimator, chosen on principle and reported alongside the other two. Added criteria A7–A9
4	"The engine is validated"	Validation used a **global** case database with **different inputs** (γ, c', φ', β, H, r_u) from the field score sheet (6 scores)	No dataset pairs field scores with outcomes	A8 (score tool) and A9 (Indonesian spatial) marked **PENDING**

#	Earlier claim	What was wrong	Evidence	Fix
5	P_f is a probability for QRA	Raw Monte Carlo P_f on real slopes is badly calibrated: worse than a constant guess	Brier 0.341 (no-skill 0.25), ECE 0.331. Slopes with $P_f \leq 0.1$ failed 22%; $P_f > 0.9$ failed only 57%	New criterion **A7 = FAIL**
6	"Exact Sobol' weights are objective"	Weights move with the class-to-parameter mapping	Soil weight 0.18–0.37; vegetation 0.07–0.28 across 7 variants	Report weight ranges; language changed
7	"Bishop matches Taylor within 0.2%"	The 45° benchmark (3.7% error) had been **left out**	V2 now includes 45°	Fixed
8	Case-history AUC set next to published susceptibility-map AUCs	Different tasks; not comparable	—	Comparison removed
9	ACADS 1(a) reference "0.99" stated as fact	Value recalled from memory	—	Marked "recalled, unverified"; reference ~1.00 (error 0.9%)
10	Case study "Class IV: urgent action"	Model output presented as a finding	Unvalidated (A8); RI class limits were designed for the weighted index, not for a probability	Labelled as model output; caveat added
11	Bootstrap CIs	Case-level resampling ignores near-duplicate groups	Group bootstrap 0.832–0.896 vs case bootstrap 0.834–0.894: practically identical	Group bootstrap adopted

3. Data problems that limit any validation

From `results/audit/audit_summary.json`, using the Ma et al. (2022) database:

- **Removed rows.** 627 raw rows; **52 removed** for $\gamma < 10$ kN/m³ (physically impossible). 56% of the removed rows are failures, so removal slightly changes the class balance.
- **Extreme parameter values.** 20 cases have $\varphi' = 0$, 52 have $c' = 0$, and 33 have $r_u \geq 1$.
- **Duplicates and label noise.** The 575 kept cases form 466 near-duplicate groups. **10 groups contain the same slope labelled both stable and failed.**
- **Physics and labels disagree. 150 of 288 "stable" slopes have Bishop FS < 1**, and **78 of 287 "failed" slopes have FS ≥ 1**. Many entries are rock or open-pit slopes with back-analysed strengths.

What this means:

- The database tests a classifier of *reported outcomes*, not the infinite-slope field model.
- Any AUC on it is an upper-level indication, not a field accuracy.

4. Validation bias: measured

All on **all 575 cases**, grouped 5-fold cross-validation × 5 repeats (`results/audit/cv_estimators_all.csv`):

Model	AUC per-fold mean	AUC single-pass	AUC 5-repeat avg	ECE per-fold	ECE single-pass	Sens. / Spec. @0.5
A Equal-weight score (not AHP)	0.579	0.581	0.583	0.104	0.074	0.59 / 0.54
B Physics only (Bishop FS)	0.648	0.645	0.646	0.094	0.054	0.47 / 0.66
E Logistic regression	0.735	0.731	0.734	0.110	0.063	0.63 / 0.70
D Hybrid, fully monotone	**0.863**	**0.860**	0.865	0.116	**0.077**	**0.81 / 0.80**
C Unconstrained boosting	0.906	0.906	0.918	0.093	0.056	0.85 / 0.83

- **Unconstrained learner (C).** C is more accurate on this database, but significantly so because of non-physical patterns: D − C = −0.052, CI −0.075 to −0.031. C is reported as a reference only, not adopted.
- **Domain choice.** Across **48 alternative domain filters**, physics-only AUC ranges **0.64–0.78** and the hybrid **0.84–0.91**. The filter used earlier was not the most favourable, but physics-only results depend strongly on it.
- **Leakage.**
- Random cross-validation inflates the hybrid's AUC by 0.014.
- Grouping duplicates up to 8× more coarsely leaves the hybrid at 0.858–0.868, so residual leakage is small.

- The unconstrained model was more sensitive to leakage.
- **Operating point (ethics).**
- At threshold 0.5 the hybrid **misses 19% of failed slopes**.
- To catch 90% of failures the threshold must drop to 0.27, and specificity falls to 0.69.
- For life-safety screening, report recall and choose thresholds by the cost of a missed failure.

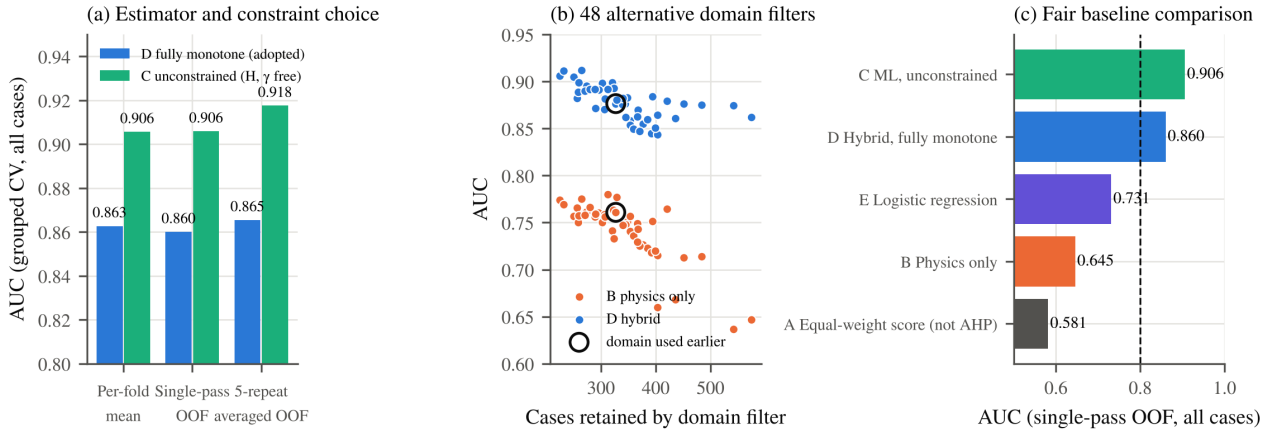


Figure 1. Reported AUC depends on the estimator and on whether γ and H are constrained (a); physics-only and hybrid AUC under 48 domain filters (b); fair comparison of all five models on all cases (c).

5. How Monte Carlo affects the model

Monte Carlo has **two roles**. They behave very differently.

5.1 Inside the score model (field tool)

Each score class becomes a distribution of physical parameters. Monte Carlo samples it N = 400 times and returns P_f = P[FS < 1].

Effect	Size	Source
Sampling noise at N = 400 (case study)	s.d. of P_f = **0.017** (0.051 at N = 50; 0.002 at N = 20,000)	`mc_sample_size.csv`
c′–φ′ correlation (ρ = 0, −0.5, −0.7)	Weights change ≤ 0.01; case P_f 0.854–0.862	`mc_assumption_sensitivity.csv`
COV × 0.5 or × 1.5	Weights change ≤ 0.02; case P_f 0.850–0.872	same
Class definitions (v1 uniform vs v2 literature)	Soil weight **0.37 → 0.19**, vegetation **0.11 → 0.26**	same
Root cohesion × 0.4 (lower-bound sensitivity)	Vegetation weight **0.26 → 0.07**; case P_f 0.86 → 0.93	same
Observer scoring error (20% chance per score of ±1 class)	90% interval of case P_f: **0.41–0.94**	`scoring_uncertainty_case.csv`

Conclusions:

1. Monte Carlo noise is negligible at N = 400.
2. The distributional details Monte Carlo makes explicit (correlation, COV) barely matter.
3. The **class definitions and root-cohesion ranges dominate** the weights, so that is where the remaining judgement sits.
4. **Scoring error dominates everything**: it is about 10× larger than Monte Carlo noise.

5.2 On real slopes (does Monte Carlo add predictive power?)

Predictor (all 575 cases)	AUC	Brier	ECE
Deterministic FS < 1 (yes/no)	0.604	0.397	0.397
−FS (ranking)	0.650	—	—
Monte Carlo P_f (c′ COV 0.45, φ′ 0.10, ρ −0.5)	0.633	0.341	0.331
Monte Carlo P_f, low COV	0.626	0.365	0.370
Monte Carlo P_f, ρ = 0	0.630	0.332	0.317

Predictor (all 575 cases)	AUC	Brier	ECE
Hybrid with ln FS vs hybrid with P_f as feature	0.862 vs 0.862		

Conclusions:

1. Monte Carlo turns a yes/no FS verdict into a graded output. That improves Brier score over the binary verdict (0.341 vs 0.397).
2. It does **not** discriminate better than FS itself.
3. Its probabilities are **not calibrated** frequencies.
4. Its value is transparent uncertainty bookkeeping. It becomes a real probability only after calibration against outcomes (e.g. isotonic/Platt on held-out data).

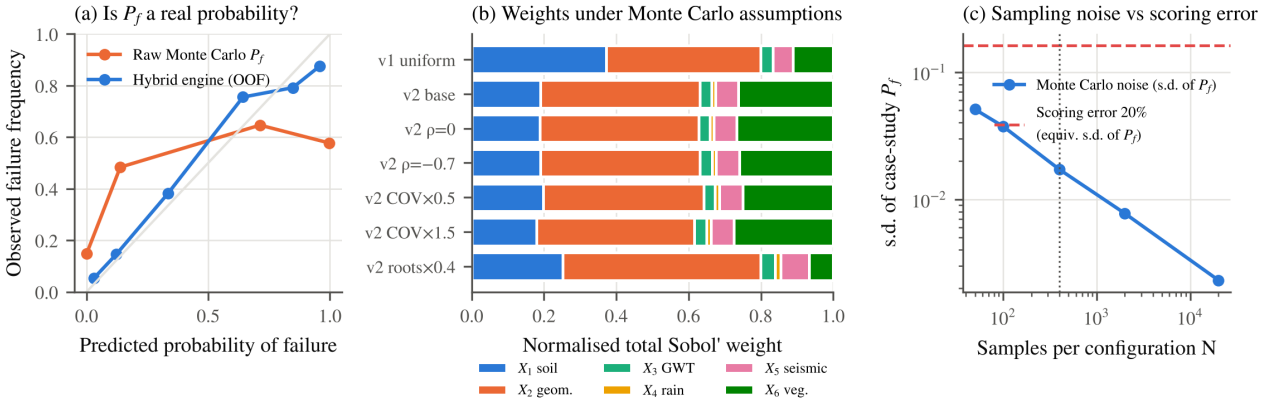


Figure 2. Calibration of raw Monte Carlo P_f and of the hybrid engine on real slopes (a); normalised Sobol' weights under seven Monte Carlo assumption sets (b); Monte Carlo sampling noise versus a 20% scoring error (c).

6. How the score is taken

1. **Assign a class to each variable.** The field engineer uses the written class table (`landslide_risk/parameters_v2.py`, PDF Table 1). For example, a 45° slope → geometry class 4 (45–55°).
2. **Look up P_f.** The six scores index one row of a **pre-computed table of 15,625 P_f values** (`results/engine/full_factorial_v2.csv`). That value is HI. **No weights are applied**: each variable's influence is already inside the physics.
3. **Compute VI.** $VI = (0.667 \cdot s_{distance} + 0.333 \cdot s_{vegetation} - 1)/4$, where $s_{distance}$ comes from the settlement-distance class (80 m → 4). These two weights are still AHP weights.
4. **Classify.** $RI = \sqrt{(HI \cdot VI)}$, then the class limits 0.2 / 0.4 / 0.6 / 0.8.

Case study: scores 3-4-4-4-1-2 give HI = 0.86, VI = 0.58, RI = 0.71, Class IV.

- **With scoring error.** If each score has a 20% chance of being one class off, the result is Class IV with probability 0.87, Class III 0.10, Class V 0.02 and Class II 0.01. At a 10% error rate, P(IV) = 0.95.
- **Weighted vs physics hazard.** Over the 15,625 combinations, the slides' weighted HI and the physics P_f rank configurations similarly (Spearman $\rho = 0.76$). They give the **same RI class in only 45%** of combinations, with physics higher in 35%. Without outcome data, neither can be called correct.

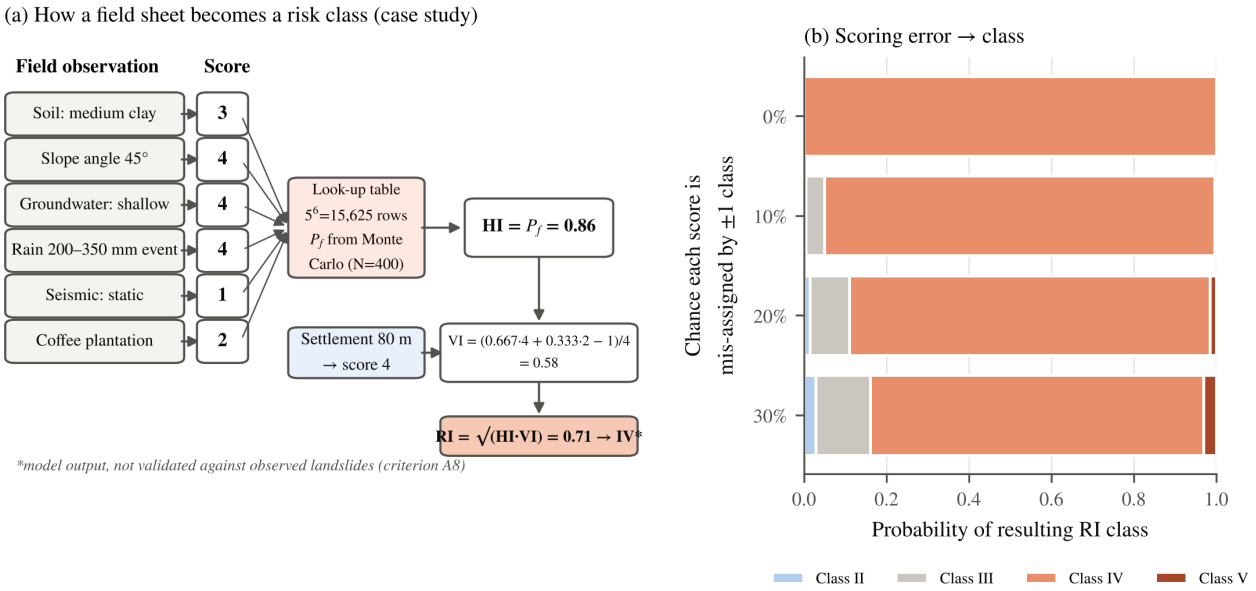


Figure 3. From field observations to scores, the P_f look-up table, HI, VI and the risk class (a); probability of each class for the case study under scoring error (b).

7. Acceptance protocol after the audit

See `results/engine/acceptance.md`.

Gate	Pass	Fail	Pending	Notes
Verification (V1–V6)	6	0	0	V1, V3, V6 are analytic. V2 now includes 45°. V4 reference is recalled
Numerical (N1–N4)	4	0	0	N1 is analytic. N2, N3 thresholds are project-set
Validation (A1–A9)	6	**1 (A7)**	**2 (A8, A9)**	A1 uses all cases and a group bootstrap

8. Conditions for ethical use

1. **Research and screening only.** Until A8 and A9 pass, outputs must not be the sole basis for evacuation, relocation or land-use restrictions. Every Class IV–V result needs review by a geotechnical engineer.
2. **Report the cost of errors.** Always report recall and false-negative rate alongside AUC, and set thresholds by the cost of a missed failure.
3. **Publish what you assumed.** Publish class tables, distributions, code and data with any map. State which numbers are *validated*, which are only *verified*, and which are *assumptions*.
4. **Don't oversell the weights.** Never present Sobol' weights as assumption-free; show their range under alternative class definitions.
5. **Calibrate before calling it a probability.** Calibrate P_f against outcomes before using it in quantitative risk assessment.

9. What would make the prediction trustworthy

1. **Spatial validation (A9).** Use the Samodra (2024) Cyclone Cempaka inventory with spatially blocked cross-validation. Use BNPB InaRISK as the official baseline.
2. **Field-score validation (A8).** Record field scores at mapped landslides and at stable control slopes. This is the only way to test the tool that would actually be used.
3. **Probability calibration (fixes A7).** Calibrate P_f on held-out outcomes, then report β_R.
4. **Inter-observer study.** Have several engineers score the same slopes, to measure the real scoring error rate that Section 6 had to assume.